



DEFECTS

Physical breakage of fragments from the tablet edge.



CHIPPED

CAPPED



Horizontal separation of the tablet's top/bottom crown



CORE TECHNOLOGIES



SAFER HEALTHCARE | BETTER QUALITY

ABOUT US

We are a team of final year undergraduate students in Computer & Systems Engineering (2022-2026 batch) at NED University of Eng & Tech.

Under the guidance of our advisor, Dr. Syed Zaffar Qasim, we aim to bridge the gap between academic research & industrial applications.

TEAM MEMBERS

MANAHIL EJAZ - CS22011
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NED
UNIVERSITY OF
ENGINEERING & TECHNOLOGY



DEPARTMENT OF COMPUTER SYSTEMS
ENGINEERING

IN-LINE

TABLET QUALITY CONTROL

USING COMPUTER VISION

*Real-Time Automated Inspection for
Pharmaceutical Precision*

SUPERVISOR: SYED ZAFFAR QASIM

SYSTEM BENEFITS & KEY FEATURES



95 % HIGH ACCURACY



REAL-TIME INSPECTION



REDUCED HUMAN ERROR & RELIABLE QUALITY



LOGGING & REPORTING



DEFECT CLASSIFICATION



INDUSTRY CHALLENGE

Traditional quality control relies heavily on manual inspection, which is time-consuming, labor-intensive, and prone to human error.



THE RISK

Sampling-based inspection methods do not ensure complete coverage, increasing the risk of defective products reaching the market and causing potential recalls.



OUR SOLUTION

We propose an automated, real-time inspection system using computer vision that ensures 100% tablet coverage directly on the production line.

SYSTEM PIPELINE



1. IMAGE CAPTURE

- 1080p webcam with ring light
- Mounted on stand above conveyor belt



2. TABLET TRANSPORT

- Custom conveyor belt
- DC motor ensures smooth, continuous movement



3. IMAGE PREPROCESSING

- Resize, normalize, enhance
- Prepare image for model input



4. MODEL INFERENCE (RASPBERRY PI 4)

- YOLOv8-seg (Instance Segmentation)
- Detects & segments defects with pixel-precise masks



5. DETECTION OUTPUT

- Defect class & mask
- Confidence score
- Pass / Fail decision



6. DATA STORAGE (INFLUXDB)

- Results & metadata stored in InfluxDB
- Optimized time-series logging



7. DASHBOARD (GRAFANA)

- Real-time monitoring
- Trends, alerts, analytics
- Accessible from web